Class 29 fractal structure

Activity

Teams 3 and 4: Post photos of your work to the course Google Drive today. Include words, labels, and other short notes that might make those solutions useful to you or your classmates. Find the link in Canvas.

Questions

1. (An interesting geometric structure. Example 4.9, Alligood et al) Consider the tent map $x_{n+1} = T_3(x_n)$, defined as

$$x_{n+1} = \begin{cases} 3x_n, & x_n \le \frac{1}{2} \\ 3(1-x_n), & \frac{1}{2} \le x_n. \end{cases}$$

   This is a slope-3 tent map. Consider it to be defined on the whole real line. Let C be the set of points on the real line whose orbits do not diverge to $-\infty$. This is the set of initial conditions where points in the orbit never become negative for T_3.

   The set C has an interesting (fractal) structure.

   (a) Sketch the map.

   (b) Initial conditions outside of $[0,1]$ all have orbits that eventually diverge.

   Convince yourselves that the interval $(1/3, 2/3)$ are the only points that leave the interval $[0,1]$ under a single iteration of T_3 (note that $1/3$ and $2/3$ both stay in the interval).

   Sketch the two line segments that are still in consideration for potentially being in C. Call this set C_1.

   (c) What points will leave $[0,1]$ under two iterations of T_3? Again sketch the line segments that are still in consideration for staying in the interval. Call this set C_2. Is the set of points closed or open? (i.e. do the intervals contain their endpoints or not?)

   (d) Can you generalize this to sketch C_3 (three iterations)? What do you think will happen with k iterations?

   (e) What points seem to be in C?

2. The middle-thirds Cantor set, C, is formed by starting with $S_0 = [0,1]$ and repeatedly removing the open middle third from every interval that remains. Let S_n be the set that remains after n steps, and let $C = S_\infty$.

   (a) Sketch S_0, S_1, S_2, and S_3.

   (b) Describe how S_{k+1} is built from scaled copies of S_k.

   (c) How many intervals are in S_k? What is the length of each interval?

   (d) What is the total length of the set S_k?

   (e) What does this suggest about the total length of the Cantor set?

3. We removed intervals of total length 1 to construct the Cantor set, and the set that remains has total length 0.

   How does this set still contain many points?

(a) At each step of the construction, a point in the Cantor set lies either in a left subinterval or a right subinterval. Use 0 to indicate left and 1 to indicate right. This gives each point an **itinerary** consisting of an infinite sequence of 0s and 1s.

What point is encoded by the itinerary $0111\ldots$ (where the 1s continue to repeat)?

(b) An infinite itinerary is associated with a nested sequence of intervals drawn from $S_1, S_2, \ldots$. Sketch the first three nested intervals associated with $0111\ldots$.

Each infinite itinerary determines a unique point in the Cantor set.

(c) Points in the interval $[0, 1]$ can also be represented by infinite binary expansions $0.a_1a_2a_3\cdots$. Does this suggest that the Cantor set has as many points as the interval $[0, 1]$?

(d) Is every point in the Cantor set an endpoint of one of the intervals in some S_n?

Consider the itinerary $010101\ldots$, *with* 01 *repeating.*

4. A self-similar set is made of m scaled copies of itself, each reduced by a factor of r. Its **similarity dimension** d satisfies

$$m = r^d.$$

(a) Using $r = 3$, find the similarity dimension of:

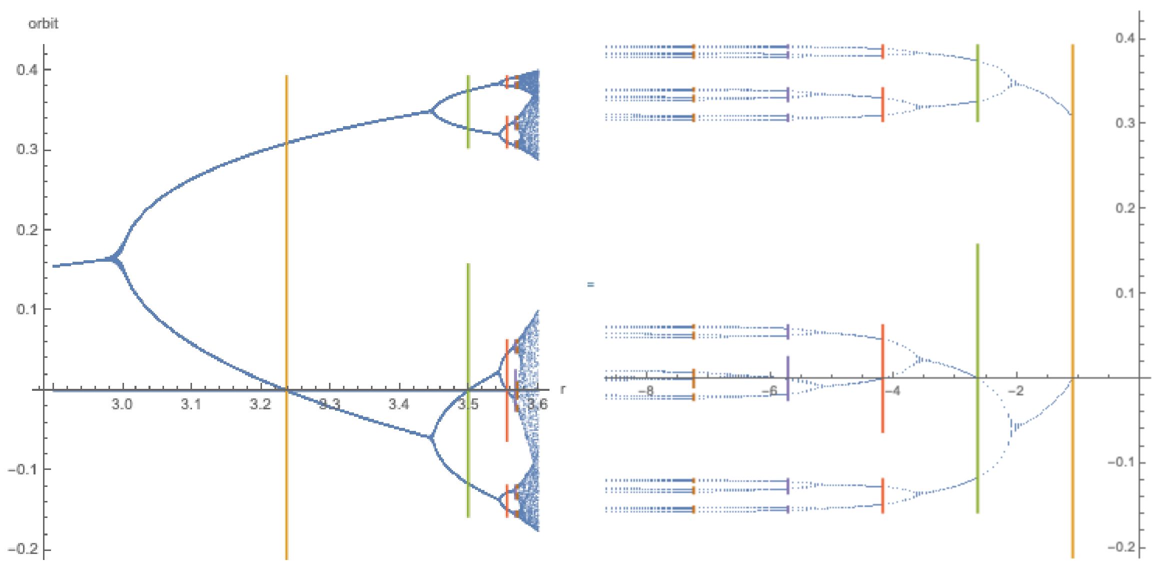
- the line segment $[0, 1]$,
- the middle-thirds Cantor set,
- the square $[0, 1] \times [0, 1]$.

Scaling by $r = 3$ *replaces a square by a smaller square with sides* $1/3$ *the original.*

(b) Compare the dimension of the Cantor set to the dimensions of the line segment and the square.

5. Based on the diagram on the right below, how does the series of period doubling bifurcations in the logistic map lead to a Cantor-like, fractal structure (a "topological" Cantor set)?

In a topological Cantor set the gaps are various sizes and the set is not strictly self-similar.



On the left is the logistic map orbit diagram (for a logistic map centered about 0). On the right, I am showing r on a log scale based on the distance to r_∞, the limiting r value associated with the period-doubling cascade.